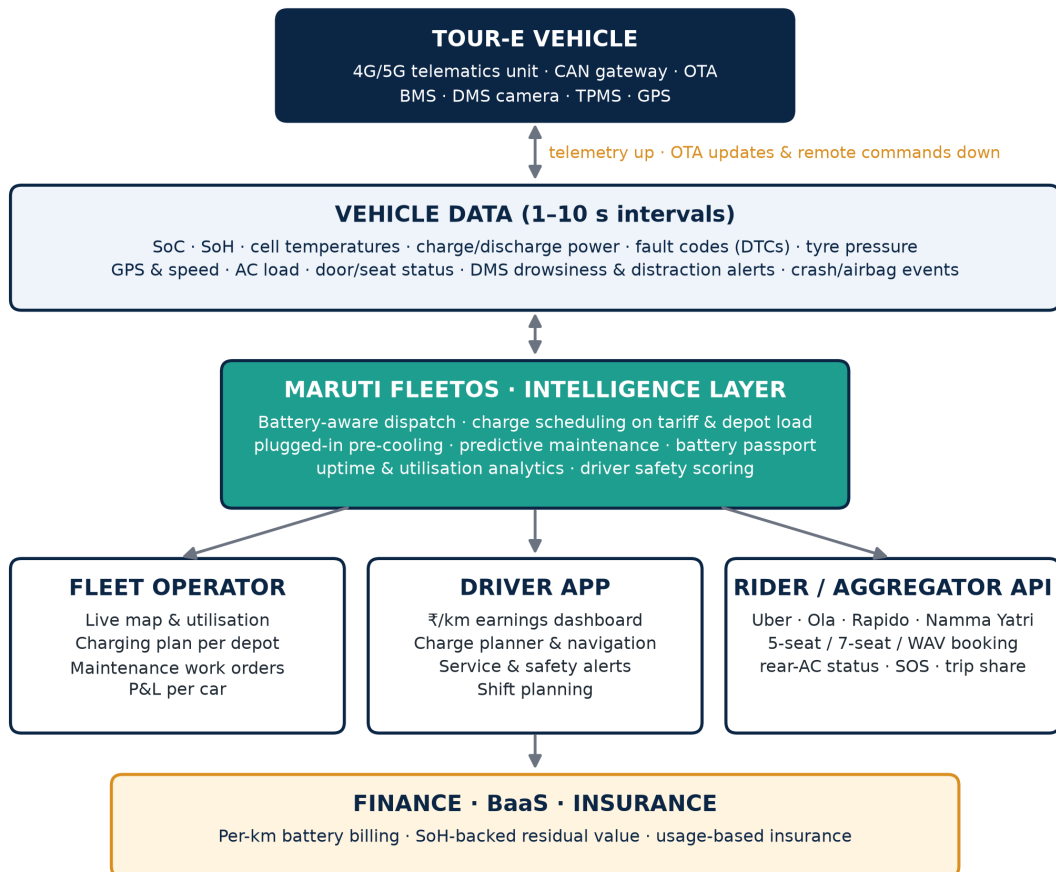


FleetOS Architecture

The connected layer: telematics, fleet and battery management, apps and data
How vehicle data becomes lower cost per km

FleetOS is the connected layer that turns each Tour-e into a managed, bankable asset. It links the vehicle, the fleet operator, the driver, ride-hailing platforms and the BaaS financier through one data backbone. Its job is simple: **more paid kilometres per day at a lower ₹/km.**

SoH + GPS + energy + duty cycle → intelligent dispatch → charging optimisation → predictive maintenance → **lower TCO**



Maruti FleetOS architecture. Data flows up from the vehicle; OTA updates and remote commands flow down.

1 · Vehicle data

Signal	Frequency	Used for
State of charge (SoC), energy used per km	Every 10 s while driving	Battery-aware dispatch, range prediction, charge planning
State of health (SoH), cell temperatures, cycle count	Every trip / every charge	Battery passport, BaaS residual value, heat protection
Charge sessions (location, kWh, power, tariff)	Per session	Depot load management, energy cost per km
GPS position, speed, harsh braking	Every 1–5 s	Dispatch, ETA, driver safety scoring, SOS location
Fault codes (DTCs), tyre pressure, brake wear	On event / daily	Predictive maintenance, service booking
DMS alerts (drowsiness, distraction)	On event	Real-time driver alerts; fatigue-aware shift planning
Cabin: AC load, door and seat status	Per trip	Pre-cooling, 5/7-seat availability for riders
Crash / airbag events	On event	Automatic emergency call and fleet alert

2 · Intelligence modules

2.1 Battery-aware dispatch

A trip is offered to a car only if it can finish the trip and still reach a charger with a safety reserve. Energy needed is estimated from the car's own recent Wh/km, adjusted for heat and AC load.

Dispatch rule: offer the trip if usable kWh now $\times (1 - 10\% \text{ reserve}) \geq (\text{trip km} + \text{pickup km} + \text{km to nearest charger}) \times \text{Wh/km} \times \text{heat factor}$

Worked example (45 kWh car at 30% SoC = 13.5 kWh)	Airport run	Outstation trip
Usable energy after 10% reserve	12.2 kWh	12.2 kWh
Trip + pickup + to charger	45 + 8 + 10 = 63 km	90 + 8 + 10 = 108 km
Energy needed (150 Wh/km $\times$ 1.1 heat factor)	10.4 kWh	17.8 kWh
Decision	Dispatch	Offer to a car with more charge; send this car to charge

So what: no low-SoC car is dispatched a trip it cannot finish, which removes stranded cars and cancelled trips.

2.2 Charge scheduling and depot load

- Plans each car's overnight charge to finish just before its next shift, filling the lowest-tariff hours first under time-of-day tariffs.
- Caps total depot load at the connection limit, so a depot needs a smaller grid connection and pays lower demand charges.
- Uses public DC only when a shift needs it, keeping ~80% of energy at the depot tariff (~₹9/kWh versus ~₹20/kWh public).
- Pre-cools the cabin and battery while the car is still plugged in, so the first trip starts cool without using battery energy.

2.3 Predictive maintenance

Component	Signals watched	Action
Battery pack	Cell temperature spread, voltage imbalance, SoH trend	Balance, limit fast charging, schedule inspection
Thermal system	Coolant temperatures, pump current, AC pressure	Service before summer peaks
Tyres and brakes	TPMS, regen share, braking events	Rotate or replace at the next depot visit
12 V system and electronics	Fault codes, 12 V battery voltage	Pre-emptive replacement to avoid no-start
Suspension	Vertical acceleration, road-quality maps	Inspect bushes and dampers on rough-road duty

2.4 Battery passport

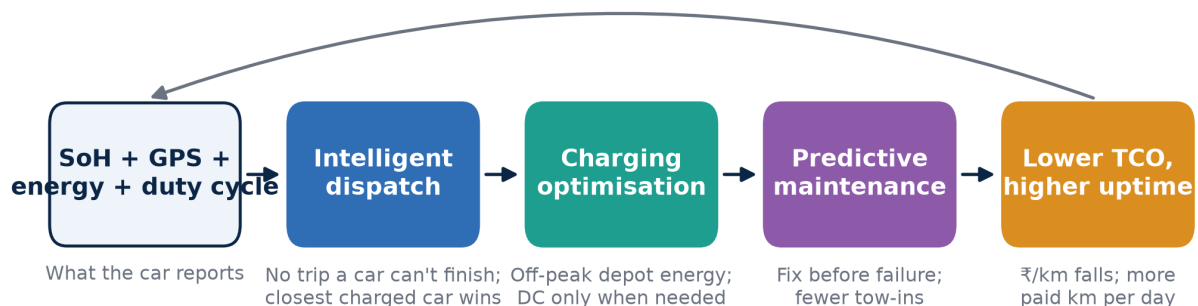
A lifetime record of each pack (cell origin, SoH history, cycles, temperature exposure and charging mix). It lets the BaaS provider price the residual value of the pack, supports second-life use and recycling under India's Battery Waste Management Rules, 2022, and gives used-car buyers proof of battery condition, which protects resale value.

3 · Apps and interfaces

User	What they get
Fleet operator	Live map and utilisation, charge plan per depot, maintenance work orders, P&L per car, driver safety scores
Driver app	₹/km earnings dashboard, charge planner with nearest available charger, service and safety alerts, shift planning
Rider (through aggregator API)	Choice of 5-seat, 7-seat or WAV; rear-AC status; SOS and live trip sharing
Aggregators (Uber, Ola, Rapido, Namma Yatri)	Open API for vehicle availability, SoC-based trip eligibility and seat configuration
Finance / BaaS / insurance	Per-km battery billing, SoH-backed residual value, usage-based insurance pricing

4 · How the data lowers TCO

Data from every trip improves the next dispatch, charge plan and service schedule



FleetOS lever	Effect on the business case
Battery-aware dispatch	Fewer cancelled trips and stranded cars; higher paid km per day
Depot charge scheduling	Keeps the blended tariff at ~₹11.2/kWh (80% depot); the model shows that 50% public DC would raise ₹/km from 7.91 to 8.45
Predictive maintenance	Protects the ₹0.55/km maintenance assumption and uptime
Battery passport and SoH data	Supports the 20% residual value of the BaaS vehicle and the 25% pack residual used to price BaaS at ₹2.0/km
Driver safety (DMS, SOS)	Lower accident risk and insurance cost; rider trust

5 · KPIs we will track

KPI	Target	Why it matters
Vehicle uptime	≥95% of scheduled shifts	Every lost shift costs ~200 km of revenue
Paid km per car per day	200 (owner-driver) · 250 (fleet)	Drives ₹/km and payback
Share of energy from depot charging	≥80%	Main lever on energy cost
Trips cancelled for low charge	~0	Battery-aware dispatch
Unplanned downtime per car per year	Falling year on year	Predictive maintenance
Pack SoH at 4 lakh km	≥70%	BaaS economics and residual value

6 · Security, privacy and compliance

- Personal data processed under the Digital Personal Data Protection Act, 2023: consent for driver and rider data, purpose limitation, data stored in India.
- Vehicle tracking and emergency button as required for public service vehicles (AIS-140).
- Signed OTA updates, encrypted telematics and role-based access for operators, platforms and financiers.

7 · Rollout

Phase	FleetOS scope
2026-28 · PROVE	FleetOS and fleet-BaaS pilot on e VITARA with 2-3 operators in Delhi NCR, Bengaluru and Mumbai; log heat and duty-cycle data
2028-31 · LAUNCH	Tour-e ships FleetOS-ready; aggregator APIs live; battery passport for every pack
2031-35 · SCALE	Depot hubs into Tier-2; airport swap pilots; AI dispatch across platforms